Developing a Natural Language Processing Pipeline to Detect Climate Nihilism

CE 395-3 / CS 396-5 / CS 496-5 / CE 495-3

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Our generation is increasingly culturally and socially nihilistic, which has also pervaded our interest and involvement in advocacy on sociocultural issues, including but not limited to engagement with climate advocacy. Climate nihilism involves an acceptance of the end of the world and/or the belief in the futility of climate action. This project is interested in creating a data scraping/NLP pipeline that would examine attitudes and trends for climate nihilism over time and different social media platforms. Detecting nihilism itself poses interesting opportunity for intersectionality psychological/linguistic/sociological research with machine learning/NLP, and figuring out how to determine both computationally/psychologically the existence of a nihilist attitude from a text excerpt is a key component of the project. Data will be scraped from social medias such as X, Reddit, or Instagram. The final product will be a dashboard where users can explore a "nihilism score" and its correlation with other features.

CCS Concepts: • **Computing methodologies → Machine learning**; **Supervised learning**.

Additional Key Words and Phrases: Machine learning, natural language processing, sustainability, nihilism, climate nihilism

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1 Aims and Objectives

Our fundamental objective is to develop a natural language processing (NLP) pipeline capable of scraping, detecting, and quantifying "climate nihilism" in social media text. The aim in doing so is to better understand the trends in climate nihilism as a practice, which is to understand one of the social bottlenecks that may or may not be hindering climate action. Our concrete objectives are as follows:

- Define a linguistic/psychological framework for what denotes "climate nihilism" vs. other behaviors, such as climate denial, general negativity, etc.
- Develop a data scraping pipeline to gather text data from platforms like Reddit and X
- Develop a data cleaning and tokenization pipeline to prepare data for NLP
- Train, test, and tune a transformer-based language classifier such as ClimateBERT
- Build a dashboard/separate visualizations plotting trends in climate nihilism in a clear, understandable format

2 Background

As the world becomes increasingly digital and more and more communication is relegated to online platforms, particularly social media platforms such as Reddit or X (formerly known as Twitter), so too has the landscape around advocacy shifted. It is increasingly important to be able to parse and analyze social media trends and posts in order to better understand how advocates, detractors, and the general public are mobilizing and organizing, including trends in popular and extremist attitudes. The study of climate change communication similarly has increasingly focused on digital tools [Falkenberg et al. 2022]. Notable trends in climate change communication on digital platform have occurred over the last decade, and computational methods have been employed to study such trends (also known as climate informatics). For example, computational studies in climate informatics have previously tracked the development of "coordinated and well-funded countermovement of climate sceptics" using text mining techniques [Boussalis and Coan 2016],or the impact of corporate funding on discourse using mined text and network data [Farrell 2016].

Communication found on digital platforms is largely text-based, and there is grounds for using linguistic or semantic context found in media to examine climate change discussions. For example, Soutter and Mottus found in their 2020 study that the framing of climate change as "climate change" made certain groups of people more likely to believe in the phenomenon as opposed to when climate change was referred to as "global warming" [Alistair Raymond Bryce Soutter 2020]. The way information is presented online may have a significant effect on how consumers of content engage with the information and may contribute to different emotional attitudes and action taken by any media consumers. Falkenberg et al's study in 2022 studied the discussion around the United Nations Conference of the Parties on Climate Change (COP) using Twitter data and linked increases in "ideological polarization" with a "fourfold increase" in right-wing activity, and how the prevalence of different themes and dynamics in Twitter climate discussion have reflected

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growing tensions in climate discussion and "'climate contrarian' views" [Falkenberg et al. 2022].

To parse digital discourse, which comes in vast amounts and is largley unstructured, researchers have previously employed natural language processing (NLP) techniques such as sentiment analysis to quantify users on metrics such as "positivity," "negativity," etc. Initial approaches included simple rule-based models, such as VADER in 2014 [Hutto and Gilbert 2014], to classify the sentiment of tweets; despite being a relatively simple model, VADER already outperformed individual human raters. Other research applied this work and linked mood on social media with decision-making, such as Twitter mood with stock market actions [Bollen et al. 2011]. Previous examples of using sentiment analysis to examine climate advocacy also exist, such as the work of Cody et al in 2015 to aggregate climate sentiment and examine trends that contribute to happiness/unhappiness related to climate news [Cody et al. 2015]. Subsequent work expanded into more nuanced NLP tasks, including stance detection in microblogging environments (Twitter being an example) [Mohammad et al. 2016] and the development of comprehensive sentiment analysis frameworks tailored specifically for environmental media coverage [Patronella 2021]. Most past work has created models that are excelling at identifying basic emotions such as positivity, negativity, and neutrality. However, there has been a consistent deficit in the granularity of past models/research in attempting to examine and capture more implicit and/or more complex psychological/emotional states.

Simultaneously as digital tools have developed, sociological and psychological research has identified a growing phenomenon which has been termed "eco-anxiety," which characterizes a pervasive sense of doom and hopelessness regarding the future, and which is particularly common among younger demographics, such as gen Z and gen alpha [Hickman et al. 2021]. We will refer to this phenomenon in our proposal more descriptively as "climate nihilism." Within the broader field of clinical NLP, strides have been made in detecting nuanced psychological states such as depression, trauma, and generalized anxiety directly from social media text. This has been achieved using established psychometric dictionaries like LIWC [Pennebaker et al. 2015] as well as predictive machine learning models capable of identifying early warning signals of mental distress [De Choudhury et al. 2013; Shah SM 2024]. Despite these parallel advancements in climate-related sentiment analysis and clinical NLP, there seems yet to be research or projects that link the two ideas together to address the specific psychological barrier of climate nihilism using computational methods.

The recent advent of transformer-based architecture and deep bidirectional models, such as BERT (Bidirectional Encoder Representations from Transformers) [Devlin et al. 2018], has revolutionized text classification by capturing deep contextual relationships in language. Further models based on BERT have been developed and trained to be adapted to specific domains; one example that might be particularly salient for this project is ClimateBERT [Webersinke et al. 2021], which additionally demonstrates just how effective pretraining deep learning models on domain-specific "environmental corpora" is for accurately parsing nuances within climate-related text. This proposed project seeks to employ the effectiveness of relatively new Transformer-based NLP methodologies to bridge the existing gap between psychological NLP methodologies discussed previously and developing more advanced climate informatics. The goal for this project is to provide a novel framework for quantifying climate nihilism, a relatively new and unstudied psychological dimension related to climate inaction, using BERT technology. Ideally, this project will contribute to better understanding a novel barrier to global sustainability efforts.

3 Challenges and Alignment to Class

Some challenges that we anticipate in undertaking this project include the inherent subjectivity and ambiguity involved in defining "nihilism". It is unclear how "nihilistic" attitudes are expressed linguistically and it may be even more difficult to define nihilism computationally. Distinguishing between some social media user expressing conflating emotions such as sarcasm, frustration, or humor and expressions of genuine nihilism/belief in futility will require careful research and development of the appropriate features, as well as thorough analysis of our dataset and potentially manual data labeling..

We also anticipate challenges with gathering and using data. For one, restrictions on social media APIs (especially X/Twitter and Reddit) are hurdles for gathering data. Fortunately, there are open-source datasets such as ClimateMNIST that have already scraped data that we can use to supplement our scraped data for training purposes. Additionally, social media data is very noisy. There might be significant class imbalance in the data since truly nihilistic tweets may be few and far between compared to expressions of other emotions in user tweets/posts. We can mitigate this class imbalance with sampling techniques during model training such as bootstrapping from the minority class data or using effective classification evaluation metrics.

Regarding alignment with this course (CE 395-3 / CS 396-5 / CS 496-5 / CE 495-3), this project aligns with the course's inherently interdisciplinary nature by combining computational, linguistic, and psychological research dimensions. Additionally, this project is a clear undertaking of the course's focus on applying computing for sustainability-oriented applications - we are applying the computing methodology of NLP to understanding a novel barrier to sustainability (climate nihilism). By quantifying this barrier, our tool provides actionable data for climate communicators and policymakers, addressing the human element required to make sustainability solutions more viable and/or more effective.

4 Outcomes and Deliverables

(1) A cleaned dataset of social media excerpts tagged for climate nihilism, which can serve as a benchmark for future sociolinguistic research
(2) A Python codebase for text preprocessing, feature extraction, and classification, alongside the final Transformer-based model for nihilism classification
(3) Deployment of an interactive dashboard with dynamic data visualizations allowing users to filter data by date, platform, and any additional features the team wishes to include (TBD)
(4) A final report/presentation detailing the methodology, model evaluation metrics, and a synthesis of our findings

5 Project Plan and Timeline

Team member roles/responsibilities:

- Data engineer [Josh Lukin]: ETL - data scraping, cleaning, and loading
- ML engineer 1 [Jinxi Zhang]: model training and data classification
- ML engineer 2 [Annie Liu]: model tuning and evaluation
- Full-stack engineer [Madeleine Worrall]: full-stack and visualization

(Note - roles may be flexible and adjust as needed based on domain/technical knowledge, as discussed in the team contract, but this division of responsibility is what we've initially devised)

Timeline:

- Week 4 and 5

  [Entire team] Conduct research into linguistic and/or psychological markers of nihilistic thinking

  [Entire team] Decide which social media platforms we want to focus on, and which one to start with specifically

  [Data engineer, full-stack engineer] Set up developer accounts for Reddit/X/other social media (?team will discuss further) APIs

  [Data engineer, full-stack engineer] Build initial data scraping script for at least one platform

  [ML engineer 1, ML engineer 2] conduct research into model types and different existing transformer-based models, evaluate which one we would like to go with

  (Entire team) Work on midterm video
- Week 6

  [Data engineer] Conduct data cleaning and text preprocessing (tokenization, stop-word removal, etc...) and create a reproducible/rerunnable pipeline

  [ML engineer 1] filter and classify data from ClimateMNIST and begin training on sample data

  [ML engineer 2] evaluate initial/baseline performance of model and work with ML engineer 2 to fine-tune classification metric

  [ML engineer 2] request access to Quest HPC

  [Full-stack engineer] begin working on dashboard (can probably work with dummy data at this point?), designing architecture and UI components of web application
- Week 7 and 8

  [Data engineer] continue working on data scraping pipeline, develop pipeline scraping from a second social media platform for comparison

  [ML engineer 1] continue training on sample data and optimizing, additionally run training on scraped data from [data engineer]

  [ML engineer 2] use Quest to devise optimal hyperparameter configuration

  [Full-stack engineer] continue working on dashboard and use test results from the model to start building interactive visualization components
- Week 9

  [Data engineer] stress test pipelines and connect the pipeline to the training model to automatically generate results

  [ML engineer 1] run final test runs and iterations of hyperparameter tuning

  [ML engineer 2] develop final evaluation metric and conduct inference on the model to evaluate overall performance and extract insights from model architecture

  [Full-stack engineer] finalize work on the dashboard and work on deployment/refining visual data representations
- Week 10 (largely meant as a buffer week for debugging code, addressing deployment issues, etc)

  [Entire team] work on pulling together final report/analyses

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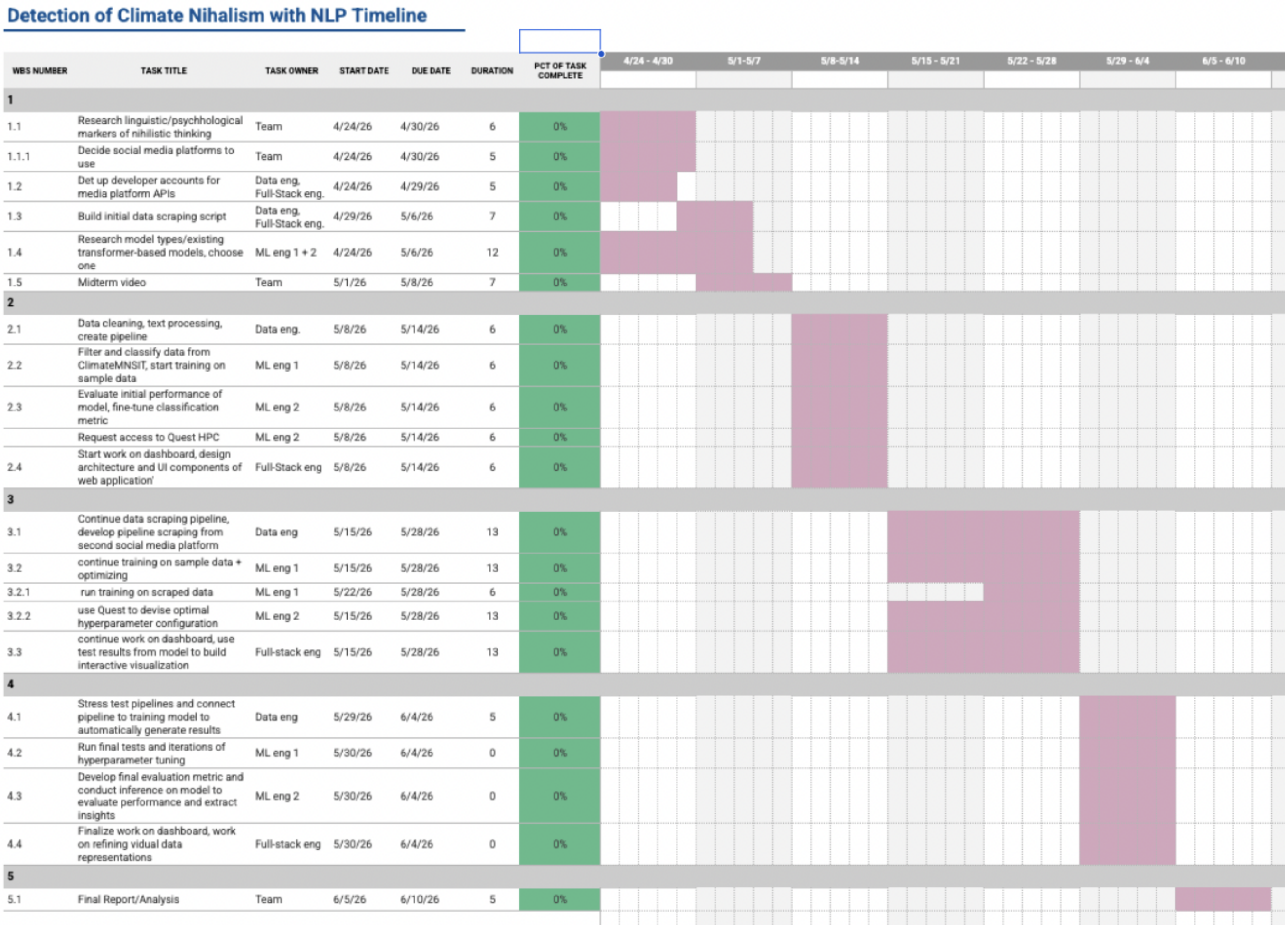
6 Appendix



Detection of Climate Nihalism with NLP Timeline

WBS NUMBER	TASK TITLE	TASK OWNER	START DATE	DUE DATE	DURATION	PCT OF TASK COMPLETE
1						
1.1	Research linguistic/psychhological markers of nihilistic thinking	Team	4/24/26	4/30/26	6	0%
1.1.1	Decide social media platforms to use	Team	4/24/26	4/30/26	5	0%
1.2	Det up developer accounts for media platform APIs	Data eng, Full-Stack eng.	4/24/26	4/29/26	5	0%
1.3	Build initial data scraping script	Data eng, Full-Stack eng.	4/29/26	5/6/26	7	0%
1.4	Research model types/existing transformer-based models, choose one	ML eng 1 + 2	4/24/26	5/6/26	12	0%
1.5	Midterm video	Team	5/1/26	5/8/26	7	0%
2						
2.1	Data cleaning, text processing, create pipeline	Data eng.	5/8/26	5/14/26	6	0%
2.2	Filter and classify data from ClimateMNSIT, start training on sample data	ML eng 1	5/8/26	5/14/26	6	0%
2.3	Evaluate initial performance of model, fine-tune classification metric	ML eng 2	5/8/26	5/14/26	6	0%
	Request access to Quest HPC	ML eng 2	5/8/26	5/14/26	6	0%
2.4	Start work on dashboard, design architecture and UI components of web application'	Full-Stack eng	5/8/26	5/14/26	6	0%
3						
3.1	Continue data scraping pipeline, develop pipeline scraping from second social media platform	Data eng	5/15/26	5/28/26	13	0%
3.2	continue training on sample data + optimizing	ML eng 1	5/15/26	5/28/26	13	0%
3.2.1	run training on scraped data	ML eng 1	5/22/26	5/28/26	6	0%
3.2.2	use Quest to devise optimal hyperparameter configuration	ML eng 2	5/15/26	5/28/26	13	0%
3.3	continue work on dashboard, use test results from model to build interactive visualization	Full-stack eng	5/15/26	5/28/26	13	0%
4						
4.1	Stress test pipelines and connect pipeline to training model to automatically generate results	Data eng	5/29/26	6/4/26	5	0%
4.2	Run final tests and iterations of hyperparameter tuning	ML eng 1	5/30/26	6/4/26	0	0%
4.3	Develop final evaluation metric and conduct inference on model to evaluate performance and extract insights	ML eng 2	5/30/26	6/4/26	0	0%
4.4	Finalize work on dashboard, work on refining vidual data representations	Full-stack eng	5/30/26	6/4/26	0	0%
5						
5.1	Final Report/Analysis	Team	6/5/26	6/10/26	5	0%

To see the Gantt chart in Google Sheets, please click the link:

https://docs.google.com/spreadsheets/d/1yu7yLcg0VUUOt6nSMYUCdkeUzDn2YqXmWKWXNRZ1XkI/edit?usp=sharing

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